



Term Project Stage 3

Team 04

ADA University

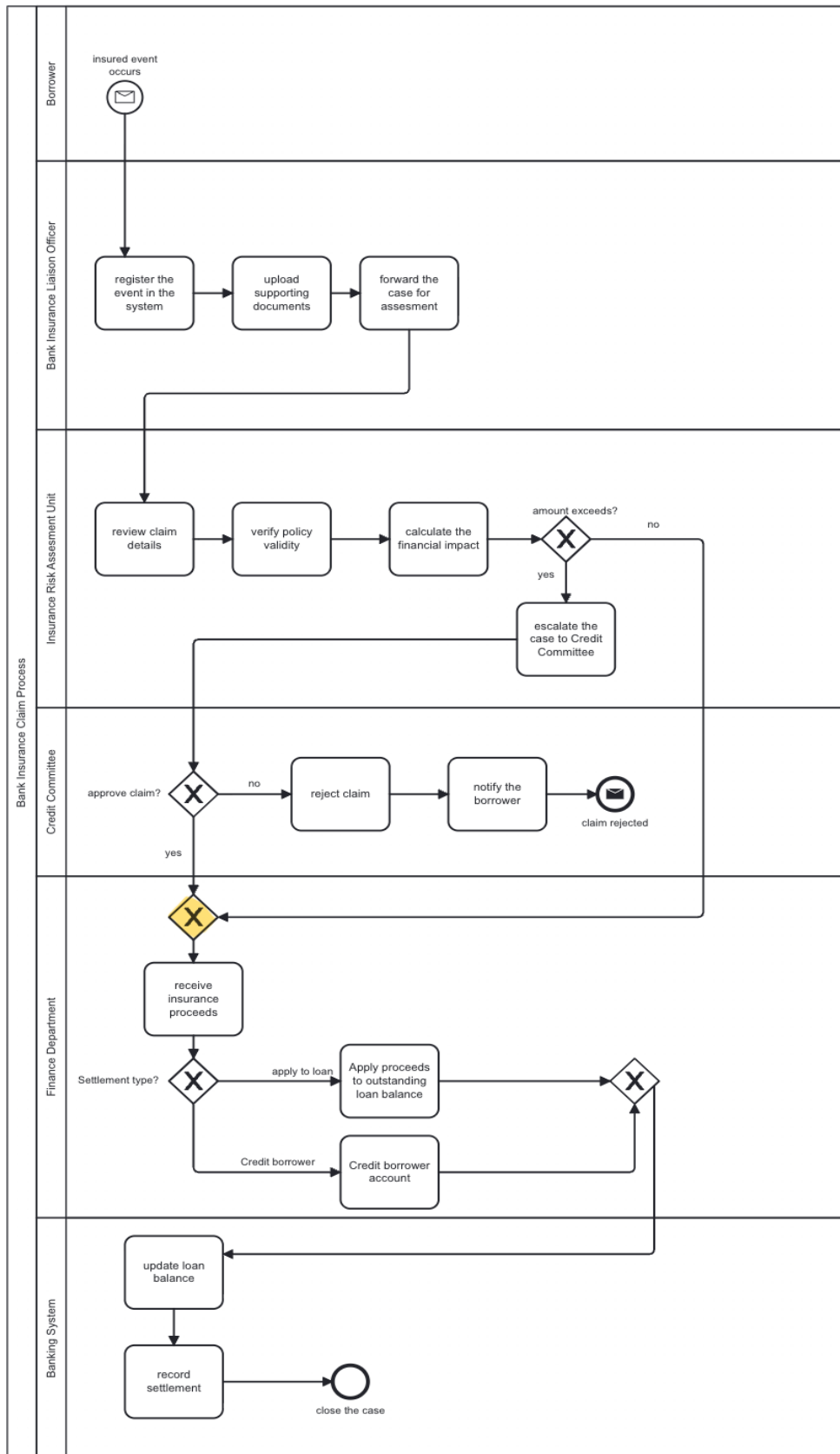
Spring 2026 Business Process Modeling for IT Solutions (INFT-3610 - 20301)

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# AS-IS Process Model



## Qualitative Process Analysis

### Value-Added Analysis

Value-added analysis is a method where each of the activities within a process are classified according to whether or not they add value to the customer or to the business. Each activity in the AS-IS model has been broken down and assigned to one of the three categories:

- VA - Value Adding. An activity directly provides a value or satisfaction to the borrower.
- BVA - Business Value Adding. An activity is necessary for the operation of the business or for regulatory compliance and the customer would not normally pay for this activity.
- NVA - Non-Value Adding. An activity does not fall into either the above and is regarded as waste to be reduced.

### Value Classification Table

|   | Activity                         | Category   | Justification                                                                                                                                                                                                                            |
|---|----------------------------------|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Register the event in the system | <b>BVA</b> | Generates the official case file which is used for all downstream processes and for audit. The borrower doesn't directly benefit from the registration-the added value only comes into play later if the claim ever comes to settlement. |
| 2 | Upload supporting documents      | <b>BVA</b> | An internal administrative step. This step is more about the bank's need for a complete file; the borrower has already lodged their notification.                                                                                        |
| 3 | Forward the case for assessment  | <b>NVA</b> | A pure step where responsibility is handed from one unit to the next (Liaison Officer to Insurance Risk Assessment Unit). The claim is                                                                                                   |

|    |                                            |            |                                                                                                                                                                                                                                                                                      |
|----|--------------------------------------------|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    |                                            |            | neither assessed nor changed in any way at this step; there is no decision made.                                                                                                                                                                                                     |
| 4  | Review claim details                       | <b>VA</b>  | This is the first time the borrower's claim has actually been properly looked at and assessed, and the loan officer decides whether it meets the policy conditions and is therefore a valid claim. The immediate benefit for the borrower lies here.                                 |
| 5  | Verify policy validity                     | <b>VA</b>  | This confirms the presence of a live insurance policy at the time of the claim. This step protects the borrower against fraudulent refusal.                                                                                                                                          |
| 6  | Calculate the financial impact             | <b>VA</b>  | This step calculates the value of the claim; it therefore has direct relevance to the amount of money the borrower receives and also influences if the claim need to go to Credit Committee.                                                                                         |
| 7  | Escalate the case to Credit Committee      | <b>NVA</b> | Triggered on a value limit of a claim, and sends the claim up to the next authority in the process -Adds nothing analytically, just moves to a higher level of internal sign off                                                                                                     |
| 8  | Approve or reject claim (Credit Committee) | <b>BVA</b> | Mandatory internal control step required as per bank's internal governance policy and by regulatory provisions to manage high value claims -Essential for the bank's risk management framework, however the borrower does not pay the bank specifically for this layer of assessment |
| 9  | Reject claim                               | <b>BVA</b> | Formally closing a rejected case- This is a mandatory process and legal step, preventing future disputes/challenges and necessary for compliance and security of the bank, but adds no benefit.                                                                                      |
| 10 | Notify the borrower                        | <b>VA</b>  | Only step in the rejection process which leads to direct borrower communication and informs them of their outcome. Direct communication to the borrower affects their ability to make further choices.                                                                               |
| 11 | Receive insurance proceeds                 | <b>BVA</b> | The step where funds come in to the bank's treasury (from the insurer) -The borrower does not get immediate benefit here, as the funds have been received by the bank, not the borrower yet.                                                                                         |

|    |                                            |            |                                                                                                                                                                                                                                                                             |
|----|--------------------------------------------|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12 | Apply proceeds to outstanding loan balance | <b>VA</b>  | These steps reduce the loan amount owed by the borrower with funds received from the insurer-This is one of the two key outcome branches of this process.                                                                                                                   |
| 13 | Credit borrower account                    | <b>VA</b>  | These steps send money directly to the borrower's account on occasions when direct transfer is called for-This is the other key outcome branch of this process.                                                                                                             |
| 14 | Update loan balance                        | <b>BVA</b> | This is purely a bookkeeping entry by the bank to reflect that settlement has been completed within the bank itself. The bank does not gain any additional benefit other than that reflected by step 12/13 and it is a legal accounting requirement for reporting purposes. |
| 15 | Record settlement                          | <b>BVA</b> | This step essentially close the case within the system after everything is done-This process simply signifies that the case is finished, it does not benefit the borrower, it is a regulatory requirement for audit.                                                        |

### Distribution Summary

| Category                           | Count | Percentage |
|------------------------------------|-------|------------|
| <b>Value-Adding (VA)</b>           | 6     | 40%        |
| <b>Business Value-Adding (BVA)</b> | 7     | 47%        |
| <b>Non-Value-Adding (NVA)</b>      | 2     | 13%        |

As we can see, the proportion of value creating activities (40%) is quite high for a financial service process. This is due to the tangible nature of an insurance claims process. So in many of the processes, actual assessment and decision making takes place. We can identify the two NVA activities – passing for assessment and raising to Credit Committee, these are internal routing steps and do not involve any value transformation and are our targets for automation/ elimination. The 47% BVA proportion is driven by governance and control

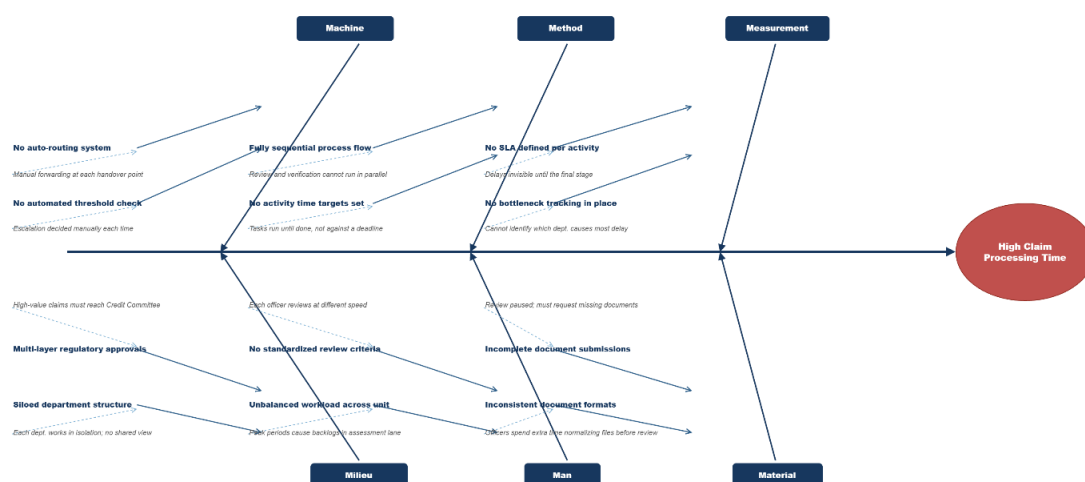
overhead which is required, but gains efficiency by performing it in parallel and using systems.

## Root Cause Analysis/ Cause Effect Diagram/Fishbone Diagram

Root cause analysis finds the actual causes behind the problems in a business process. Here, a Cause-Effect (Fishbone) diagram is constructed using the framework of 6M (Machine, Method, Material, Man, Measurement and Milieu) to pinpoint the root causes of two important issues found in the AS-IS process. For each issue, primary causes and secondary causes are separated.

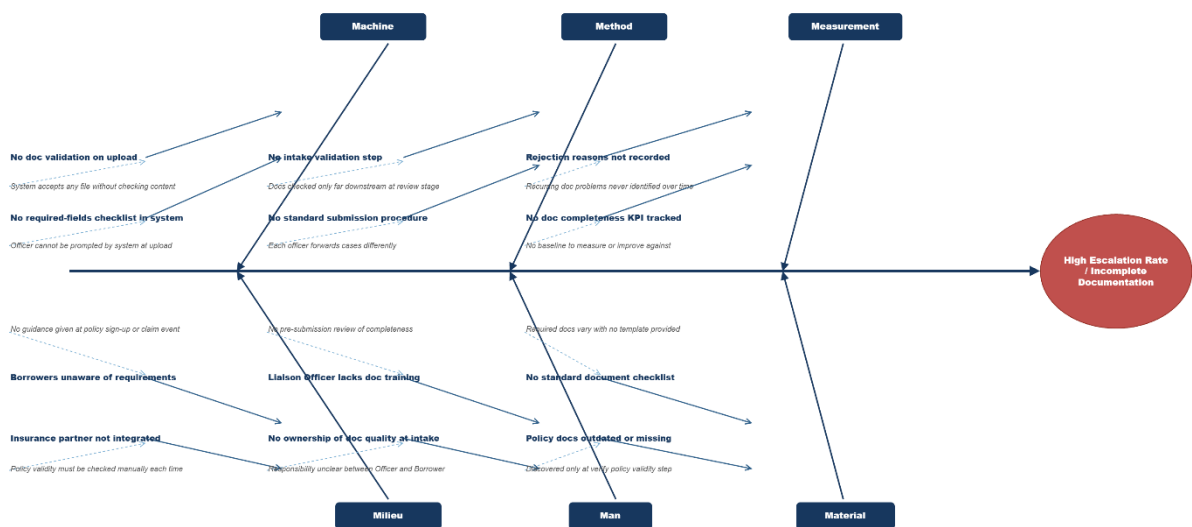
### *Issue 1: High Claim Processing Time*

The first problem of takes too long from register until settlement. Machine does not link the cases between department, Credit Committee threshold check is performed manually, Method processes all activities serially without parallelism and without specific activity time targets, Material with incompleteness documents from borrowers and variety of file format increases work, Man has no consistent assessment criterion on the insurance Risk Assessment Unit, workload isn't distributed between man, there's backlog in peak times and Measurement, there is no activity-based SLAs defined, so individual process bottlenecks remain hidden till the process is complete and Milieu due to multi-layer approval requirement for high-value claims and lack of a unified process view across the segregated departments.



## Issue 2: High Escalation Rate Due to Incomplete or Inaccurate Documentation

The second problem of too frequent rejected and referred cases, caused by documented issues found late. Machine does not perform document input check and provide required-field checklist, Method has no Intake Check activity as document check takes place late in Review Claim Details and Material, no standard checklist on documents and missing policy documents. Man, Insurance Risk Assessment Unit is not performed preliminary completeness check of documents; responsibility of the quality of document received is ambiguous and Measurement, reasons are not recorded which could show if the problem of documents is re-occurring, no KPI for document completion. Milieu, borrowers generally unaware of specific documentation requirements, and the insurance partner is not integrated into the process.



## Pareto Analysis

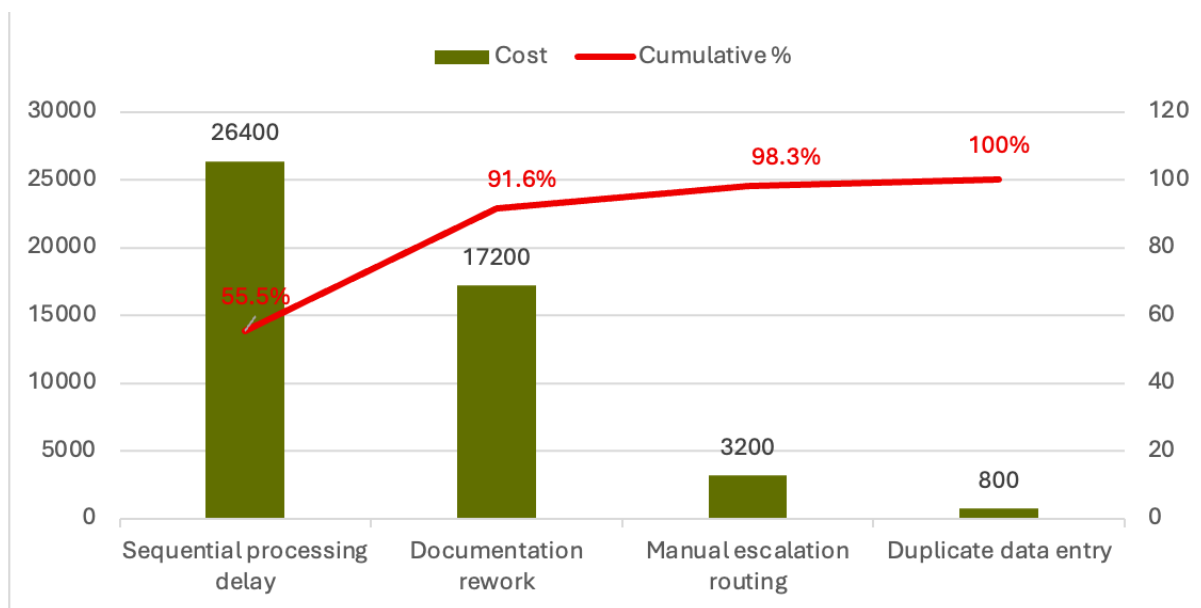
### Impact Assessment

A quantified impact assessment was carried out in order to prioritize the problems found. In all calculations below the following assumptions were kept constant: cost of an employee of 40 AZN per hour, 20 claims processed per month (this is a rather conservative calculation, representing a medium-sized bank, so could be considered safe to use), 8



working hours a day, and a 35% re-workrate on documentation. This is only to be used for prioritizing and is defined as:

|   | Issue                       | Calculation Basis                                                     | Monthly Cost      | Cumulative |
|---|-----------------------------|-----------------------------------------------------------------------|-------------------|------------|
| 1 | Sequential processing delay | 20 claims x 3.3 hrs excess wait x 40 AZN/hr x 10 staff-touches        | <b>26,400 AZN</b> | 55.5%      |
| 2 | Documentation rework        | 20 claims x 35% rework rate x 1.5 days x 8 hrs x 40 AZN/hr            | <b>17,200 AZN</b> | 91.6%      |
| 3 | Manual escalation routing   | 20 claims x 20% escalation rate x 2 hrs x 40 AZN/hr x 2 staff         | <b>3,200 AZN</b>  | 98.3%      |
| 4 | Duplicate data entry        | 20 claims x 1 hr x 40 AZN/hr                                          | <b>800 AZN</b>    | 100%       |
|   | <b>TOTAL</b>                | Assumptions: 40 AZN/hr · 20 claims/month · 8-hr day · 35% rework rate | <b>47,600 AZN</b> |            |

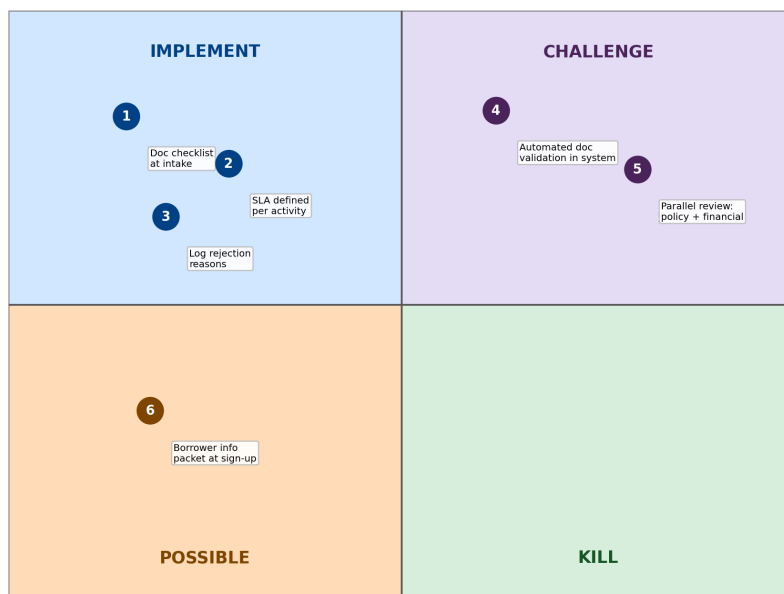


It validates the 80-20 rule since top two problem areas, sequential processing lag, and re-documentation of rework; make up 91.6% of total estimated monthly expense 47,600AZN. Hence these two problem areas will be the ones to be improved upon.

## PICK Chart

In the PICK chart, every suggested process improvement action is plotted on two dimensions; its effect on process performance (Y-axis, Low to High) and its difficulty of implementation (X-axis, Easy to Hard) and each action will be placed in one of four boxes, namely; Implement (High impact, easy implementation), Challenge (High impact, difficult implementation), Possible (Low impact, easy implementation) or Kill (Low impact, difficult implementation).

| Quadrant  |   | Action                                                                                  | Rationale                                                                                                                            |
|-----------|---|-----------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| Implement | 1 | Add document checklist upon intake                                                      | No system cost; addresses the primary root cause of Issue 2 immediately                                                              |
| Implement | 2 | Define and execute the SLA per activity                                                 | Simple policy change; will allow for instantaneous awareness of activities which are not proceeding quickly enough                   |
| Implement | 3 | Log rejection reasons into the banking system automatically                             | Low effort; will allow for measured and guided process improvements; will not allow for recurrent problems to slip through the net   |
| Challenge | 4 | Automated system build to validate documents at upload for banking systems              | High level of development required, removes the systemic cause of Issue 2 permanently, schedule for next IT cycle                    |
| Challenge | 5 | Process re-engineering to run policy validity and financial impact analysis in parallel | System modification is required and process needs to be re-engineered, drastically decreases the whole cycle time                    |
| Possible  | 6 | Deliver to borrower a pack of all relevant info upon application sign up                | Simple delivery, source of error removal and will reduce documentation error, but will not fix the inefficiencies within the process |



|   | Action                                        | X - Difficulty<br>(1–10) | Y - Impact (1–10) | Quadrant  |
|---|-----------------------------------------------|--------------------------|-------------------|-----------|
| 1 | Mandatory doc checklist at intake             | 2                        | 8                 | Implement |
| 2 | SLA defined per activity                      | 3                        | 7.5               | Implement |
| 3 | Log rejection reasons systematically          | 2.5                      | 6.5               | Implement |
| 4 | Automated doc validation in banking system    | 7                        | 8.5               | Challenge |
| 5 | Parallel review redesign (policy + financial) | 8.5                      | 7                 | Challenge |
| 6 | Borrower info packet at policy sign-up        | 2                        | 3                 | Possible  |

#### *Actions:*

Immediate execution should focus on creating a mandatory document check list for intake, define per-activity SLAs and logging of rejection reasons. Each of these 3 actions address two top cost drivers and will not require substantial system investment to implement.

Automating the document check and the redesign for parallel processing will be reserved for a future phase of improvement. An informational packet for borrowers given at policy signup would be a relatively low cost support action to decrease errors at the source.

## Quantitative Process Analysis

### Cycle Time Analysis

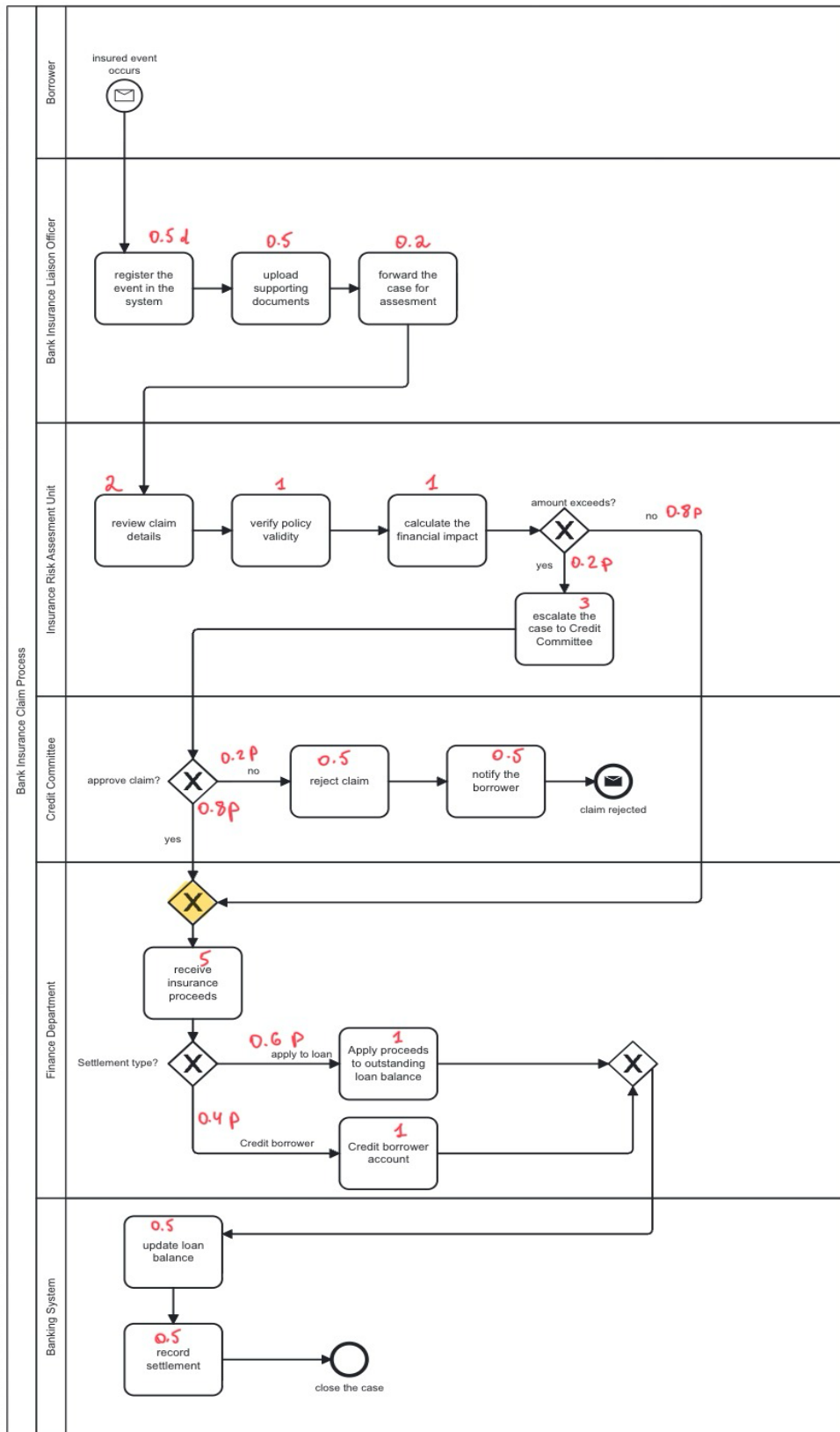
The given case did not provide the numerical durations of the activities, so logical estimation of both cycle time and processing time was made based on the nature of these tasks. In accordance with the definition used in lecture, cycle time is defined as the total elapsed time of an activity, including both actual work time and the waiting time associated with the activity. To ensure consistency of the cycle time values assigned to each activity, the following general cycle times were used: routine administrative tasks were given short cycle times, review and validation activities were assigned medium cycle times, and activities requiring approval or are dependent upon some external processes were assigned longer cycle times.

Once cycle time had been established for each activity, processing time, as the actual work time for individual activities, was determined by assuming that there would be no queuing or delay due to handovers. As an example, escalation to the Credit Committee or receipt of funds from insurance claims would have low processing times because a significant portion of their duration results from time spent waiting for an outcome as opposed to time spent performing work. Thus, these cycle time and processing time values represent analytical assumptions that will be used to create calculations for CT, TCT and CTE as per and according to the AS-IS model analysis methodology.

Table below shows the estimated cycle times and processing times for each activity, assuming that one working day equals 8 hours. Probabilities for taking each of the possible pathways from XOR gates were also logically assigned:

- Amount exceeds?: Yes = 20% (0.2); No = 80% (0.8)
- Approve claim?: Yes = 80% (0.8), No = 20% (0.2)
- Settlement type?: Apply to loan = 60% (0.6), Credit borrower = 40% (0.4)

| Activity                                   | Cycle Time (days) | Processing Time (days) |
|--------------------------------------------|-------------------|------------------------|
| Register the event in the system           | 0.5               | 0.1                    |
| Upload supporting documents                | 0.5               | 0.1                    |
| Forward the case for assessment            | 0.2               | 0.05                   |
| Review claim details                       | 2.0               | 0.5                    |
| Verify policy validity                     | 1.0               | 0.25                   |
| Calculate the financial impact             | 1.0               | 0.5                    |
| Escalate the case to Credit Committee      | 3.0               | 0.1                    |
| Reject claim                               | 0.5               | 0.1                    |
| Notify the borrower                        | 0.5               | 0.05                   |
| Receive insurance proceeds                 | 5.0               | 0.1                    |
| Apply proceeds to outstanding loan balance | 1.0               | 0.25                   |
| Credit borrower account                    | 1.0               | 0.25                   |
| Update loan balance                        | 0.5               | 0.1                    |
| Record settlement                          | 0.5               | 0.1                    |



Once the cycle times and processing times were assigned, Cycle Time (CT) was calculated in the following way:

Sequential Part Before the First Decision:

$0.5 + 0.5 + 0.2 + 2 + 1 + 1 = 5.2$  days -> used in all pathways

- Pathway A: The amount does not exceed

$$5.2 + (5 + (0.6 \times 1 + 0.4 \times 1) + 0.5 + 0.5) = 12.2 \text{ days}$$

- Pathway B: The amount exceeds & Claim is approved

$$5.2 + (3 + 5 + (0.6 \times 1 + 0.4 \times 1) + 0.5 + 0.5) = 15.2 \text{ days}$$

- Pathway C: The amount exceeds & Claim is rejected

$$5.2 + (3 + 0.5 + 0.5) = 9.2 \text{ days}$$

Since the process model does not include any AND gates or Rework Loops, only XOR gates were used in calculations.

The total Cycle Time (CT), is therefore calculated by weighted average of probabilities of XOR gates:

$$CT = 0.8(\text{Path A}) + 0.2(0.8(\text{Path B}) + 0.2(\text{Path C}))$$

$$CT = 0.8(12.2) + 0.2(0.8(15.2) + 0.2(9.2)) = 12.56 \text{ days}$$

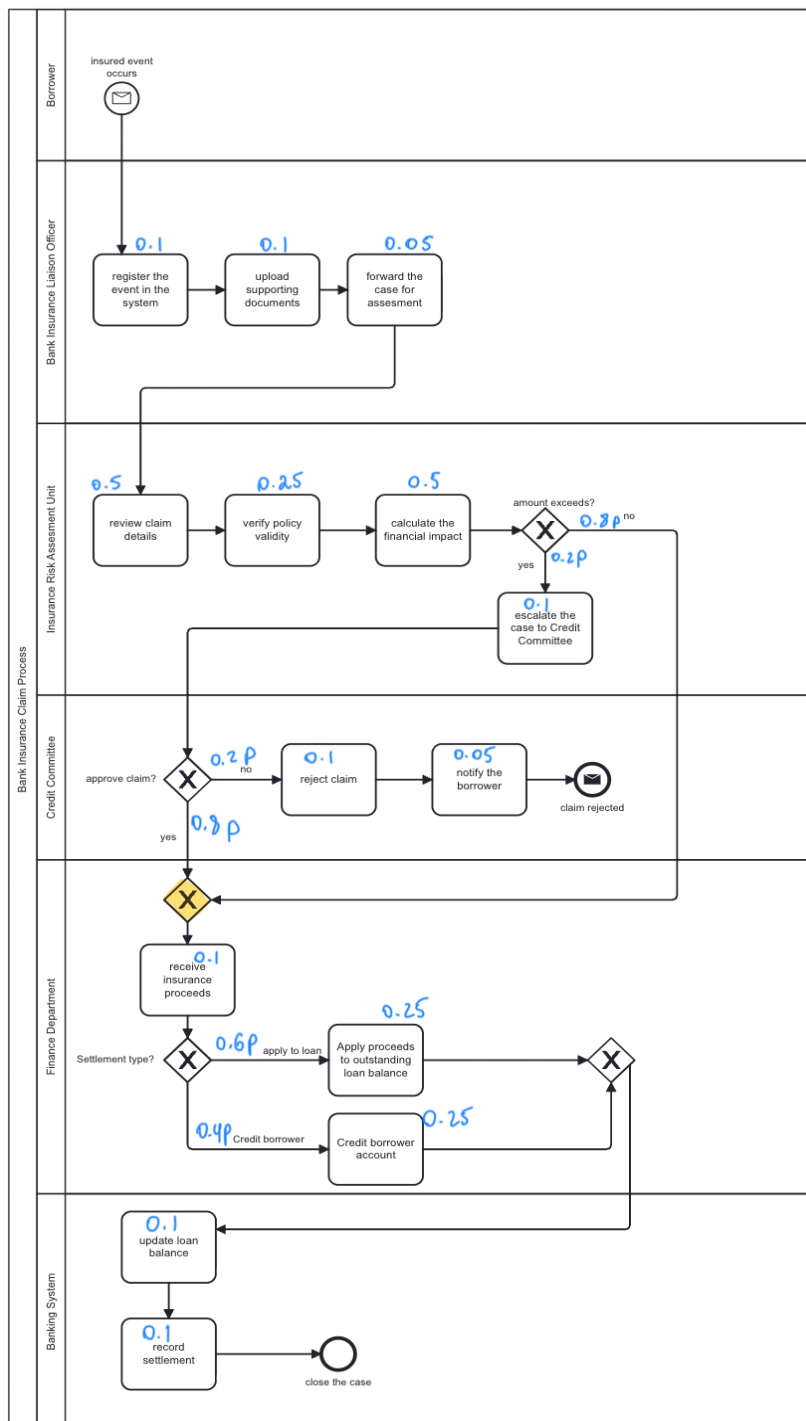
So, the overall average of the AS-IS model is 12.56 workdays.

### **Bottleneck Identification**

The aspects that have the greatest impact on the overall time required are the longest lengths of time involved and that create the longest path. Of the activities contributing to the longest total time, the most crucial is Receive Insurance Proceeds since that is an activity completed in almost every case and takes the longest time to complete at 5 days. The two other most significant contributors to the total elapsed time of the completing the process are the Escalate the Case to Credit Committee (3 days) and Review Claim Details (2 days), both of which have been identified as major contributors to total cycle time. Collectively these

tasks create the majority of bottlenecks in the AS-IS process. These bottlenecks also align with what was discussed in class regarding the definition of a critical path; that the critical path is a task sequence that delineates the cycle time for the overall process.

## Cycle Time Efficiency





Using the assigned processing times from the table above, the CTE was calculated.

Cycle Time Efficiency, (CTE) shows the amount of the total elapsed time for an entire process is used to perform actual work versus the total time spent waiting to do that work. Based on earlier modules, we calculated the Theoretical Cycle Time (TCT) using the same methodology and formulas used to calculate overall cycle time, with one exception - we substitute processing times for cycle times.

Sequential Part Before the First Decision:

$0.1 + 0.1 + 0.05 + 0.5 + 0.25 + 0.5 = 1.5$  days -> used in all pathways

- Pathway A: The amount does not exceed

$$1.5 + (0.1 + (0.6 \times 0.25 + 0.4 \times 0.25)) + 0.1 + 0.1 = 2.05 \text{ days}$$

- Pathway B: The amount exceeds & Claim is approved

$$1.5 + (0.1 + 0.1 + (0.6 \times 0.25 + 0.4 \times 0.25)) + 0.1 + 0.1 = 2.15 \text{ days}$$

- Pathway C: The amount exceeds & Claim is rejected

$$1.5 + (0.1 + 0.1 + 0.05) = 1.75 \text{ days}$$

The same structure and same sets of XOR probabilities used to compute Cycle Time are now used with Processing Times (not Cycle Times) to calculate Cycle Time Efficiency (CTE).

The overall Theoretical Cycle Time (TCT):

$$\text{TCT} = 0.8(\text{Path A}) + 0.2(0.8(\text{Path B}) + 0.2(\text{Path C}))$$

$$\text{TCT} = 0.8(2.05) + 0.2(0.8(2.15) + 0.2(1.75)) = 2.054 \text{ days}$$

The Cycle Time Efficiency is then:

$$\text{CTE} = \text{TCT} / \text{CT}$$

$$\text{CTE} = 2.054/12.56 = 0.1635 (\sim 16.35\%)$$

A lower value indicates that you spend more time waiting than working, whereas a Cycle Time Efficiency of near 1 would indicate less time waiting than performing productive work.

## **Interpretation of Results**

These results indicate that there is an imbalance of waiting vs. productive work in the AS-IS Process (12.56 total cycle days, only 2.054 days of productive work if all wait times were eliminated). The total amount of productive work is equivalent to 16.35% of the total duration of the process. In other words, 83.65% of the overall cycle time for the process consists of wait time, transfer time, approval delays, and external dependency time. From a lecture perspective, having a low cycle time to date indicates there is a significant opportunity for improvement in the process by reducing the amount of waiting, as well as the number of handovers.

This understanding is also aligned with the activity structure of the process. The longest wait times occur in the activities that consist of receiving the proceeds from an insurance company, escalating the case to the credit committee, and reviewing the claim for details. Those activities are either dependent on an external party or involve wait time within the organization. Therefore, the largest opportunity to improve the process is not created by simply having the employees that perform those activities work faster, but by reducing the amount of time that is spent in idle time, approval delays, and interdepartmental transfer time.

## **Arrival Rate ( $\lambda$ )**

The estimated arrivals were made based on what seemed like appropriate generalizations relative to a typical bank's process for handling insurance claims. Insurance claim events represent the occurrence of relatively rare events. These are triggered by insured events such as property loss/damage or death and therefore do not relate directly to the volume of regular banking transaction volumes. Because of that, while the number of customers using a bank may be high on a day-to-day basis, those same customers rarely trigger an insurance claim event. Consequently, the number of new claims entering the system at any point in time is limited.

Given this perspective, it is estimated that the bank will receive approximately 10 new insurance claims per month. As a mid-size bank with a modestly large mortgage product line, the number of insured events will be relatively low frequency. While this is simply an assumed arrival rate for analytical purposes, it should be noted that this value can be validated using internal data from the Bank if a full evaluation of this model is desired.

The monthly arrival rate can also be converted to a daily rate. In order to make this conversion, we assume a typical working month consists of 20 working days. Therefore:

$$\lambda = (10 \text{ cases / month}) / (20 \text{ working days / month}) = 0.5 \text{ cases / day}$$

Thus, under typical conditions, there is one new insurance claim entering the system every two workdays.

### **Work-In-Process (WIP)**

The WIP is the number of cases being worked on in an organization at a specific time. It can also be referred to as the work-in-progress. According to Little's Law, the WIP is equivalent to the Arrival Rate multiplied by the Average Cycle Time.

$$WIP = \lambda \times CT$$

As mentioned previously we have already determined the Arrival Rate to be 0.5 cases per day. Also, based upon our previous calculations, we have an average cycle time of 12.56 working days. Therefore:

$$WIP = \text{Arrival Rate} \times \text{Average Cycle Time} = 0.5 \text{ cases / day} \times 12.56 \text{ days} = 6.28 \text{ Cases}$$

Therefore, there will generally be around six to seven open claims simultaneously throughout the entire insurance company (from initial claim submission to final settlement).

To assess whether the calculated WIP is reasonable, we should first determine how long claims generally remain in each activity of the process. We have identified in the Bottleneck Analysis that the three longest running processes were "Receive Insurance Proceeds" (5 days), "Escalation to Credit Committee" (3 days), and "Review Claim Details"

(2 days). Claims are usually waiting in one of these three areas for a significant portion of their total elapsed time. Specifically, since many claims will wait longer than two weeks for receipt of payment from the external insurer, it is certainly possible that multiple claims may simultaneously be inactive waiting in one of these three areas while other claims are active in either a preceding or succeeding area. Therefore, an observed WIP of six to seven active cases appears to reasonably reflect both the structure and timing of the business process.

### **Verification of Little's Law**

The calculated WIP of 6.28 cases (which was close to the range of 6-7 cases) indicated that Little's Law held true for this process as it operated normally. Therefore, all assumption regarding arrival rates and cycle times were logically consistent with one another.

### **Capacity Analysis**

#### *Current Capacity Under Normal Conditions*

At present with an arrival rate of  $\lambda = 0.5$  cases per day and an average CT of 12.56 days, this process has an acceptable WIP of 6.28 cases. Therefore, this process can handle its arrival rate without an excessive amount of resource utilization of the bank employees.

A key observation to be noted is that the principal constraint in this process is the waiting period which occurs prior to receiving the service from an external entity. The Cycle Time Efficiency (CTE), 16.35%, indicates that only 2.054 days of the total 12.56 days of the cycle time constitute "productive" or "working" time. The remaining 83.65% of the cycle time represents "waiting," which includes waiting for payment from the external insurance provider and waiting for Credit Committee availability. Thus, at the existing level of arrivals, there is not full usage of bank resources, and the bottleneck is located beyond the control of the bank.

#### *Peak Period Analysis*

Peak periods, including those caused by widespread natural disasters, floods, or other events causing similar damage to many homes, result in an exponential increase in the number of insurance claims filed. Consequently, there will be an increased amount of cases arriving into the system. We have defined two possible peak period arrival rates:

- A moderate peak:

$$\lambda (\text{peak}) = 1.5 \text{ cases/day (three times the normal rate)}$$

- A severe peak:

$$\lambda (\text{peak}) = 2.0 \text{ cases/day (four times the normal rate)}$$

For each of the above peak levels, Little's Law was applied using our assumed average cycle time of 12.56 days. Then the WIP (Work-in-Process) was calculated for each scenario as follows:

- Peak Moderate WIP:

$$\text{WIP} = 1.5 \times 12.56 = 18.84 \text{ cases}$$

- Peak Severe WIP:

$$\text{WIP} = 2.0 \times 12.56 = 25.12 \text{ cases}$$

The WIP values indicate that the credit committee will experience a significant number of escalations with respect to its current workload. As indicated earlier, the credit committee is responsible for handling approximately 20 percent of escalated claims on a daily basis. If the volume of escalations were to grow by 50 percent, the credit committee would likely require additional resources to handle its growing workload. Additionally, the finance department would need to process a larger volume of incoming proceeds and settlements thereby placing additional pressure on their ability to complete their work without error. Finally, the bank insurance liaison officer would need to register and upload documentation from the mortgagees on a greater number of files concurrently than he/she

does under normal circumstances. These factors would create a bottleneck at the beginning of the process creating longer wait times and ultimately contributing to increased cycle time.

One final factor that would impact cycle time is the length of time it takes to receive payment from the insurer. Currently, the length of time is fixed at five business days and represents the greatest contribution to cycle time. While an increase in internal demand may provide some benefit by allowing the team to continue working throughout the day, it is unlikely that cycle time would shorten as a result. Furthermore, as previously discussed, if the insurer was similarly impacted by a major loss, the amount of time required to obtain payment from them could actually increase due to their inability to meet existing obligations. Overall, the combination of increased WIP and CT will create a positive feedback loop whereby higher arrival rates create more WIP leading to increased pressure throughout the organization.

### **Recommendations Based on Capacity Analysis**

Based on the information provided by the Capacity Analysis and the law of Little's, two major priority recommendations were developed. The first recommendation is that the bank has an agreement to reduce its external processing wait times by no longer than 2-3 business days through a formal negotiated service level agreement with their insurance partner. Currently the average estimated time is 5 business days which will create the greatest positive impact toward lowering cycle time. Lowering cycle time from 12.56 days to approximately 9.56 days will decrease the banks normal work-in-process from 6.28 cases to 4.78 cases, while decreasing moderate peaks work-in-process from 18.84 cases to 14.34 cases. These are the substantial reduction in overall process workload without having to make any changes internally to process operations.

The second recommendation is that the bank develop a contingency escalation plan for when it processes a large volume of requests. Under this contingency, either the Credit

Committee will meet more often or the committee can delegate authority for approvals on standard cases to a sub-committee thereby eliminating the 3 day escalation period for high volume requests. When used together, these two recommended items directly address the two largest delays in the entire process and will provide the greatest improvement in terms of both cycle time efficiency and capacity resilience.

### **Task Breakdown Table**

| Team Member                  | Contribution to project                                           | Estimated % |
|------------------------------|-------------------------------------------------------------------|-------------|
| Ulkar Ahmadli                | Quantitative Analysis: Little's Law and Capacity Analysis         | 33.3%       |
| Team leader: Seljan Musayeva | Quantitative Analysis: Cycle Time Analysis, Cycle Time Efficiency | 33.3%       |
| Farid Aslanli                | Qualitative Analysis                                              | 33.3%       |
| Lutvu Aliyev                 | Didn't join groupwork                                             | 0%          |